Code No: 156DY **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August - 2022
QUANTITATIVE ANALYSIS FOR BUSINESS DECISIONS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Elaborate on the various techniques for solving problems in Operations Research.
b) Describe the nature and scope of OR. [10+5]

2.a) Ampere Electric Company produces two products P_1 and P_2. Products are produced and sold on a weekly basis. The weekly production cannot exceed 25 for product P_1 and 35 for product P_2 because of limited available facilities. The company employs total of 60 workers. Product P1 requires 2 man-weeks of labour, while P_2 requires one man week of labour. Profit margin on P_1 is Rs.60 and on P_2 is Rs.40. Formulate a linear programming problem with the above information and solve for maximum profit graphically.
b) Explain duality principle with an illustration. [10+5]

3.a) A firm manufacturing a single product has three plants I, II and III. They have produced 60, 35 and 40 units respectively during this month. The firm had made a commitment to sell 22 units to customer A,45 units to customer B, 20 units to customer C, 18 units to customer D and 30 units to customer E. Find the minimum possible transportation cost of shifting the manufactured product to the five customers. The net unit cost of transporting from the three plants to the five customers is given below:

Plant		Customer A	**B**	**C**	**D**
	I	4	1	3	4
	II	2	3	2	3
	III	3	5	2	4

b) Write brief on application of transportation on model in OR. [10+5]

4.a) A departmental head has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. His estimates of the times that each man would take to perform each task is given below in the matrix:

	Tasks I	II	III	IV
Subordinates A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

How should the tasks be allocated to subordinates so as to minimize the total hours?
b) Examine the importance of assignment model in OR. [10+5]

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5.a) What is group replacement policy? Explain the rules governing group replacement policy.

b) What are situations which make the replacement of items necessary? [9+6]

6.a) A manual stamper currently valued at Rs.1000 is expected to last two years and costs Rs.4000 per year to operate. An automatic stamper which can be purchased at Rs.3000 will last four years and can be operated at an annual cost of Rs.3000. If money carries a rate of interest of 10 per cent per annum, determine which stamper should be purchased.

b) How do you describe the need of group replacement of low-cost items? [10+5]

7. Solve the following 3 × 5 game using dominance property. [15]

Player A strategy	Player B strategy				
	1	2	3	4	5
I	6	15	30	21	6
II	3	3	6	6	4
III	12	12	24	36	3

8. A repair shop attended by a single mechanic has an average of four customers an hour who bring small appliances for repair. The mechanic inspects them for defects and takes six minutes on an average. Arrivals are Poisson and service rate has the exponential distribution. You are required to

a) Find the proportion of time during which there is no customer in the shop.

b) Find the probability of finding at least one customer in the shop.

c) What is the average number of customers in the system?

d) Find the average time a customer spent by a customer in the shop including service. [15]

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